

Enjoy This Wireless 3D Mouse And Keyboard

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We have seen different types of human interface devices (HID), such as mouse and keyboard. In older versions of the mouse, optical sensors were used to detect movement relative to a surface, thus they required a smooth surface to function properly. They could work in only two coordinates. With advancements in technology, we are now talking about 3D user interfaces (3D UI). So, let us make a 3D HID device which

has the capability to work in all three coordinates.

In this project, we are going to make a 3D wireless HID device that will give you a whole new experience of using computers and playing online games. With this device, you will be able to control computers and smartphones by just moving your hand in the air.

This project has two units: transmitter and receiver. The transmitter

unit is to be attached to the user's hand and the receiver unit is to be attached to your computer.

This device can be very helpful in making 3D UI based systems. We can use this device in 3D designing as well. For instance, you can draw any design by moving your hand in the air. Further, you can develop the device for 3D holographic display and virtual reality (VR) that gives us a 3D UI.

The system can be further developed for differently-abled people (without hands) to enable them to use computers and smartphones. They can wear this device on their head and operate gadgets by moving their head.

To start our interesting project, we need to gather the following components:

- Arduino Pro Mini 5V, 16MHz
- Two Bluetooth HC-05 modules
- MPU6050 sensor
- Arduino Leonardo Pro Micro (ATmega32U4)
- Jumper wires
- Micro USB cable
- Four pushbutton switches

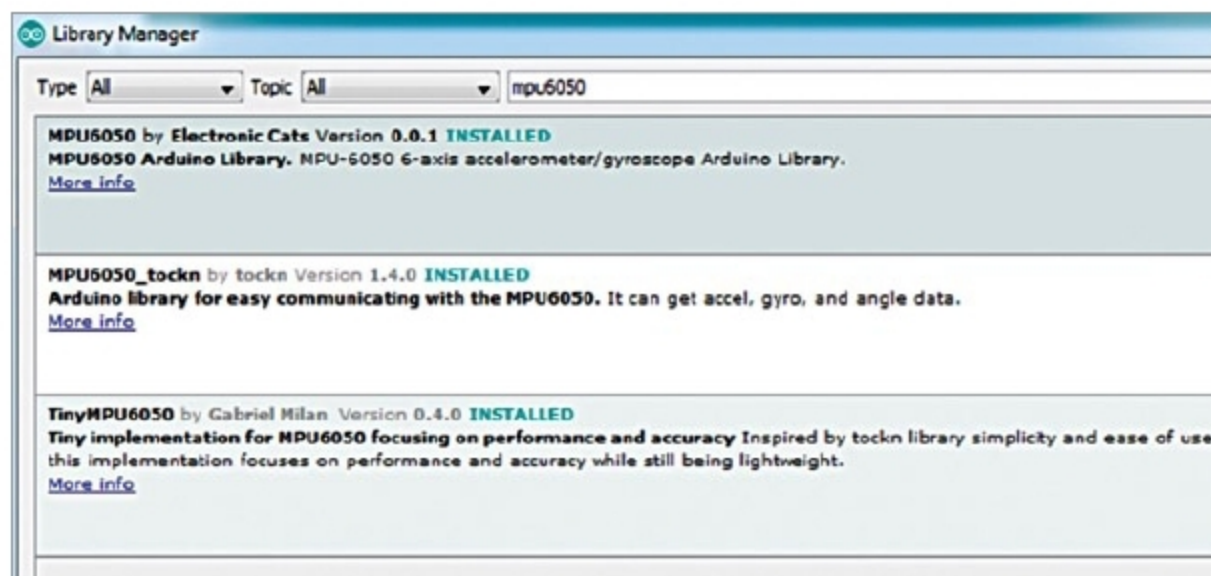


Fig. 1: Installing library



Fig. 2: Arduino Mini code set up for sensor



Fig. 3: Arduino Mini code getting MPU6050 data

Coding

First of all, we need to install an MPU6050 library. Open library manager of Arduino IDE and search for MPU6050, then install the library (refer Fig. 1). Then set one Bluetooth as a slave and another as master; follow the Bluetooth instructions on the Internet to do this.

As we have installed the library, now we can start the coding part. In the first part of coding, we will include the library for MPU6050 and set the variables for storing values of the MPU6050 sensor. After that, we can set up the sensor and Bluetooth in set up function (refer Fig. 2). I have used a baud rate of 9600 in this code.

In the second part of coding, we will send the buttons and sensors data